EXP.NO: 4	**Create a web form that accepts a user's name and age. Write a Servlet to process the form data and display it back on the browser.**

AIM:

To create a simple web application that includes a form to accept a user's name and age, and then use a Servlet to process the submitted data. The Servlet will retrieve the form data, process it, and display the values back on the browser.

ALGORITHM:

1. Input:

- The user fills out a web form with their **name** and **age** and submits it.

2. Form Handling and Servlet Processing:

1. **Create the Web Form:**
    - Create an HTML form with **two input fields**: one for the user's **name** and one for the user's **age**.
    - Add a **submit button** to the form.
    - Set the form's action attribute to the URL of the **Servlet** and the method to POST (or GET).
2. **Create the Servlet Class:**
    - Write a Servlet that processes the **form data** submitted by the user.
    - The Servlet will override the doPost() (or doGet() depending on the form method) to handle form submissions.
    - The Servlet will **retrieve the name and age** parameters from the request object using request.getParameter("name") and request.getParameter("age").
3. **Process the Data in the Servlet:**
    - Extract the name and age values submitted from the form.
    - **Validate the input** (e.g., check if the age is a valid number).
    - Prepare a **response** that will display the user's **name** and **age** back on the browser.
4. **Send Response to the Browser:**
    - The Servlet will send a **response** to the browser containing a message like:

        - "Hello, [name]! You are [age] years old."
    - The response will be displayed to the user in the browser.

5. **Deploy the Servlet:**
    - Compile the Servlet and deploy it to a **Servlet container** (like **Apache Tomcat**).
    - Ensure the web.xml file or Servlet annotations are used to map the Servlet to the URL defined in the form's action attribute.
6. **Test the Application:**
    - Open the web browser, fill in the form with the name and age, and submit it.
    - The Servlet will process the data and display the message back to the user in the browser.

HTML File (index.html):

```
<!DOCTYPE html>
<html>
<head>
  <title>User Form</title>
</head>
<body>

  <h2>User Details Form</h2>

  <form action="UserServlet" method="post">
    Name: <input type="text" name="name" required><br><br>
    Age: <input type="text" name="age" required><br><br>
    <input type="submit" value="Submit">
  </form>

</body>
```

```
</html>
import java.io.*;
import jakarta.servlet.*;
import jakarta.servlet.http.*;
import jakarta.servlet.annotation.WebServlet;

@WebServlet("/UserServlet")
public class UserServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request,
                          HttpServletResponse response)
                          throws ServletException, IOException {

        String name = request.getParameter("name");
        String ageStr = request.getParameter("age");

        try {
            int age = Integer.parseInt(ageStr);

            // Store values inside request object
            request.setAttribute("username", name);
            request.setAttribute("userage", age);

            // Forward to next page
            RequestDispatcher rd = request.getRequestDispatcher("result.jsp");
            rd.forward(request, response);

        } catch (NumberFormatException e) {
```

```
            response.getWriter().println("Invalid age entered");
        }
    }
}
```

Java File (UserServlet.java) :

```
import java.io.*;
import jakarta.servlet.*;
import jakarta.servlet.http.*;
import jakarta.servlet.annotation.WebServlet;

@WebServlet("/UserServlet")
public class UserServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request,
                          HttpServletResponse response)
                          throws ServletException, IOException {

        String name = request.getParameter("name");
        String ageStr = request.getParameter("age");

        try {
            int age = Integer.parseInt(ageStr);

            // Store values inside request object
            request.setAttribute("username", name);
            request.setAttribute("userage", age);

            // Forward to next page
            RequestDispatcher rd = request.getRequestDispatcher("result.jsp");
            rd.forward(request, response);
```

```
        } catch (NumberFormatException e) {
            response.getWriter().println("Invalid age entered");
        }
    }
}
```

Result.jsp:

```
<%@ page language="java" contentType="text/html; charset=UTF-8" %>
<!DOCTYPE html>
<html>
<head>
   <title>Result Page</title>
</head>
<body>

   <h2>
      Hello, ${username}! You are ${userage} years old.
   </h2>

</body>
</html>
```

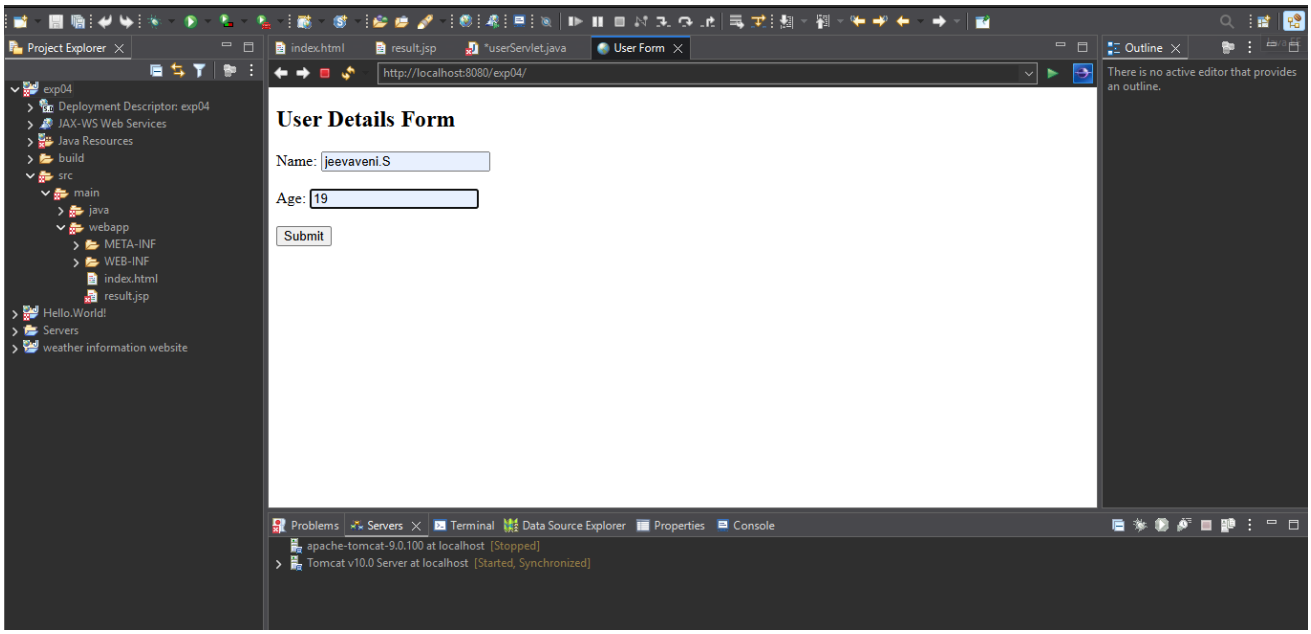
File (web.xml) :

<servlet>

```
    <servlet-name>UserServlet</servlet-name>
    <servlet-class>UserServlet</servlet-class>
</servlet>

<servlet-mapping>
    <servlet-name>UserServlet</servlet-name>
    <url-pattern>/UserServlet</url-pattern>
</servlet-mapping>
```

OUTPUT:



RESULT:

Thus A Servlet was created to process the HTML form data and display it back on the browser.